

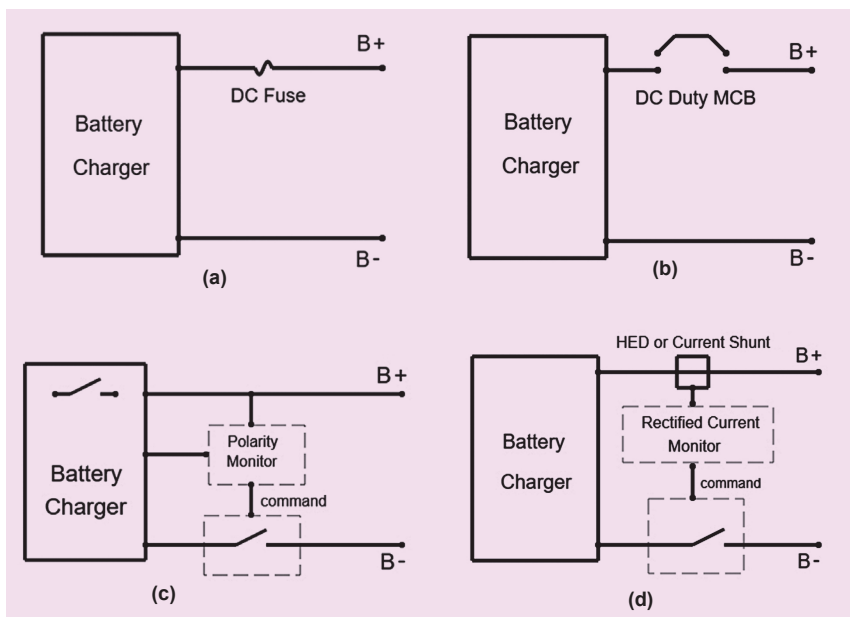


Fig. 3: (a) Fuse link on positive terminal of the charger, (b) single-pole MCB on positive terminal of charger, (c) crow-bar circuit across charger terminals to trip MOSFET switch, (d) isolated current monitor across charger terminals to trip MOSFET switch

resistance of the connecting leads. This large current flow is, most of the times, enough to damage the charger permanently.

The absolutely required protection for such unintended connection error is to clearly and legibly label the positive and negative terminals of a battery in mutually contrasting colours, such as red and blue. This visual feedback is absolutely essential in addition to the + and - marks on the battery terminals.

The next simplest mechanism to protect the charger is to install a fuse at the charger output. This fuse must be of adequate current and voltage rating, typically twice the charger's rated output current and at least twice the charger's maximum output voltage. Such a fuse is almost always guaranteed to blow up as soon as the battery is connected in reverse while the charger output supply is present. This is also the most common way to provide reverse polarity protection to chargers. This will protect the charger and, in addition, render the charger useless until the fuse is replaced with an equivalent. Since

such a fuse blow is rare, these fuse links are installed inside the charger chassis to avoid tampering, which makes the process that much more time consuming.

A much better solution is to create a circuit element that monitors the battery polarity as it is connected to the charger. The circuit could immediately disconnect one or both legs of the charger (the charger terminals) from internal electronics, if an incorrect battery polarity is detected, and would not reconnect until the correct polarity is detected. Polarity monitoring would involve terminal voltage, and for a deeply discharged battery the terminal voltage might be below the threshold, which may give incorrect results.

A further enhancement to the above protection mechanism is to include a rectified current monitor, preferably isolated from the battery supply, across a burden resistance. If the battery is connected with incorrect polarity, there shall be no current flow, signalling to the control system that the battery is not connected properly. The control system can in turn drive a solid-state

switch, such as a MOSFET placed in the path of the charger terminal, to disconnect the charger from the battery. Such a mechanism involves no blown fuses and allows the charger to be reused immediately after correcting the battery terminal polarity. Since this is a safety related mechanism, it should be preferably kept separate from the main charging controller to provide feedback to the charging system about battery connections.

Short-circuit protection

It is interesting to note that the above reverse polarity protection mechanisms, which are based on observing current through a burden resistance, can also save the charger from an output short-circuit situation. While this protection would not follow a typical I^2t curve, it still provides adequate protection to the charger against large instantaneous current flow.

Some chargers use an MCB rated for the given DC duty at the output of the charger to provide the above protections. However, the MCB is rated only for a few fault events and cannot be reused indefinitely over and over again to break the same current each time reliably. Also, general MCBs are better suited for AC currents, which contain a natural zero crossing that naturally reduces the instantaneous current flow and helps break the current flow with lesser damage to the breaker elements.

Battery over-voltage protection

In some cases, battery chargers may be subjected to battery voltages higher than their maximum output voltages. In such cases, there is a chance of reverse current flow into the charger electronics, or higher voltage stresses across the main switching elements. The most common way to protect against this is to include a diode of rated current forward biased towards the positive